THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE CHAPTER TEST SET B

Chapter 8: **Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. At a point in a medium, 50 density oscillations occur in 5 s. The frequency of the sound wave is: 1

(a) 55 Hz **(b)** 0.1 Hz **(c)** 20 Hz **(d)** 10 Hz

2. A sound wave of frequency 20 Hz has a time period of: 1

(a) 20 s **(b)** 0.03 s **(c)** 0.05 s **(d)** 0.1 s

3. A source of frequency 1000 Hz produces waves of wavelength 0.34 m in a medium. The speed of the wave is: 1

(a) 1000.34 m s^{-1} **(b)** 340 m s^{-1} **(c)** 2941.18 m s^{-1} **(d)** 680 m s^{-1}

4. Sound of frequency 500 Hz travels at 1500 m s^{-1}. Its wavelength is: 1

(a) 6 m **(b)** 3 m **(c)** 750000 m **(d)** 0.33 m

5. An echo is heard 3 s after a shout. If the speed of sound is 340 m s^{-1}, the distance of the reflecting surface is: 1

(a) 510 m **(b)** 520 m **(c)** 255 m **(d)** 1020 m

6. Thunder is heard 3 s after lightning is seen. Taking v = 340 m s^{-1}, the lightning is at a distance of: 1

(a) 510 m **(b)** 2040 m **(c)** 1030 m **(d)** 1020 m

7. Sound waves in air are: 1

(a) light waves **(b)** longitudinal waves
(c) electromagnetic waves **(d)** transverse waves

8. The audible range of a human is about: 1

(a) 20 kHz to 200 kHz **(b)** 20 Hz to 20 000 Hz
(c) 200 Hz to 2000 Hz **(d)** 2 Hz to 200 Hz

9. Loudness is measured in: 1

(a) decibel **(b)** watt **(c)** hertz **(d)** metre

10. Bats find their prey in the dark by: 1

(a) echolocation **(b)** reverberation **(c)** tone **(d)** infrasound

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 10–15 words each.

11. Define wavelength, frequency and time period of a sound wave. 2

12. Write the relation between frequency and time period. If ν = 5 Hz, find T. 2

13. State the relation between speed, frequency and wavelength. 2

14. A wave has a frequency of 200 Hz and a wavelength of 1.7 m. Find its speed. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 30–40 words each.

15. Derive v = ν λ. 3

16. Find the wavelengths for 20 Hz and 20 kHz in air (v = 344 m s^{-1}). 3

17. Lightning is seen and thunder is heard 5 s later. Find the distance (v = 340 m s^{-1}). 3

OR

How does the speed of sound vary in solids, liquids and gases? Mention the effect of temperature. 3

18. A person claps in front of a wall and hears an echo after 0.5 s. Find the distance (v = 340 m s^{-1}). 3

OR

Why is the minimum distance for an echo 17 m? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** Human hearing spans about 20 Hz to 20 kHz; take v = 344 m s^{-1}. 5

(i) Find the wavelength at 20 Hz.
(ii) Find the wavelength at 20 kHz.
(iii) Which is infrasonic and which ultrasonic?
(iv) Name one animal that detects each.

OR

(i) Derive v = νλ. 5
(ii) A wave of 500 Hz has a wavelength of 0.68 m; find its speed.
(iii) Why does the wavelength change when the wave enters another medium?

20. **Case study:** A ship sends a sonar signal and receives the echo after 5 s; v = 1525 m s^{-1}. 5

(i) Find the depth.
(ii) Why is ultrasound used?
(iii) Name the principle.
(iv) Another signal returns in 2 s; find the depth.

OR

(i) What is an echo and when is it heard? 5
(ii) Find the minimum distance of the reflector.
(iii) Distinguish echo and reverberation.
(iv) How are reverberations controlled in auditoriums?

END OF QUESTION PAPER Total: 40 Marks